

Step by Step Guide to Submit a Job

Step 1: Prepare Your Application Script (`demo.py`)

```
#!/usr/bin/env python3
print("Hello World!")
```



- Make it executable:

```
chmod +x demo.py
```



Step 2: Create a Wrapper Script (`run.sh`)

Use this script to set up your environment and stage inputs/outputs:

```
#!/bin/bash
# Activate the virtual environment
source /nethome/$USER/.venv/bin/activate

# Load any needed modules
module load python/3.10 cuda/11.7    # Example of loading a module

# Diagnostics (optional)
nvidia-smi
echo "CUDA_VISIBLE_DEVICES=$CUDA_VISIBLE_DEVICES"
echo "Running on node: $HOSTNAME"

# Prepare output directory on scratch
export SAVE=/scratch/$USER/myjob
mkdir -p $SAVE

# Execute your commands
python demo.py    # The actual task to run
```



- Make it executable:

```
chmod +x run.sh
```



Step 3: Write the HTCondor Submit File (`job.sub`)

Documentation (<https://htcondor.readthedocs.io/en/latest/users-manual/submitting-a-job.html>)

```
universe          = vanilla
#universe         = docker
#docker_image     = docker.coli.uni-saarland.de/htcondor-test:22.02-py3
initialdir        = /nethome/$USER
executable        = /nethome/$USER/projects/htcondor-test/scripts/run.sh
#comment for interactive submission
output            = /scratch/$USER/logs/htcondor-test/logfiles/run.sh.%(ClusterId).$(Year)_$(Month)_$(Day)_$(SUBMIT_TIME).out
error             = /scratch/$USER/logs/htcondor-test/logfiles/run.sh.%(ClusterId).$(Year)_$(Month)_$(Day)_$(SUBMIT_TIME).err
log               = /scratch/$USER/logs/htcondor-test/logfiles/run.sh.%(ClusterId).$(Year)_$(Month)_$(Day)_$(SUBMIT_TIME).log
request_CPUs      = 1
request_memory    = 10G
request_GPUs      = 1
requirements      = (GPUs_GlobalMemoryMb >= 16000)
+MaxWallTime      = 604800 # in seconds #For batch jobs, the default value
                        #For interactive jobs, the default
                        value is 2 hours, maximum is 24 hours
accounting_group   = pausable # <== Add this line if you want pausable jobs!

queue 1
```



- **Important Notes:**

- To use a Docker image, switch to

```
universe = docker
docker_image = docker.coli.uni-saarland.de/yourimage:tag
```



- Replace `/nethome/\$USER` and `\$USER` with your actual username.
- Make sure the log directories exist:

```
mkdir -p /scratch/$USER/logs/htcondor-test/logfiles
```



Step 4: Submit and Monitor Your Job

- Submit your job:

```
condor_submit job.sub
```



- Check job status:

```
condor_q
```



- Tail (live view) your output:

```
tail -f /scratch/$USER/logs/htcondor-test/logfiles/run.sh.$(ClusterId).$(Year)_${Month}_${Day}_${SUBMIT_TIME}.out
```



Step 5: See Outputs

- Your output ("Hello World!") will appear inside:

```
/scratch/$USER/logs/htcondor-test/logfiles/run.sh.$(ClusterId).$(Year)_${Month}_${Day}_${SUBMIT_TIME}.out
```



For anyone using coding agents, this job submission skill might be useful:
<https://gist.github.com/Yupe-Du/14aeac4c8146bc2937d7f83b923b4477>
(<https://gist.github.com/Yupe-Du/14aeac4c8146bc2937d7f83b923b4477>)

 user/cluster/job_script_example.txt  Last modified: 2026/05/15 14:36 by yudo

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